

Software Requirements Specification (SRS) For Mall Management System

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Name: Htet Aung Wai

1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for the Mall Management System (MMS). The system is designed to streamline the management of mall tenants, rental spaces, payment tracking, maintenance requests, and reporting for mall administrators and tenants.

1.2 Scope

The MMS will:

- Manage tenant information, lease agreements, and room/space details
- Track rent, utility payments, and other fees
- Handle maintenance requests and communication between management and tenants
- Provide role-based dashboards and reporting for admins, accountants, managers, and tenants
- Support both web and future mobile app interfaces
- Ensure data confidentiality, integrity, and availability in line with ISO 27001.

1.3 Definitions, Acronyms, Abbreviations

- MMS: Mall Management System
- Tenant: A business or individual renting space in the mall
- Admin: System administrator with full access
- Manager: Mall manager with limited administrative rights
- Accountant: Handles financial transactions and records

2. Overall Description

2.1 Product Perspective

The MMS is a cloud-based web application with future mobile support. It integrates with:

- Payment gateways (for online rent/utility payments)
- Document management systems (for lease agreements)
- SMS/Email services (for notifications and reminders)

2.2 Product Functions

1. Tenant Management - registration, lease management, document storage
2. Space Management - room/space details, floor plan visualization
3. Payment Management - rent, utilities, late fees, receipts
4. Maintenance Management - request submission and tracking
5. Communication System - internal messaging between management and tenants
6. Reporting & Dashboards - financial, occupancy, and maintenance reports

2.3 User Characteristics

- Admins: Full system access, configuration, user management
- Managers: Oversee tenants, spaces, and reports
- Accountants: Manage payments, invoices, and financial records
- Tenants: View their info, payments, contracts, and submit maintenance requests

(These users are just optional, I need to confirm this clearly)

2.4 Constraints

- Must support Myanmar language
- Must be cloud-deployed and accessible online
- Role-based access control is mandatory

2.5 Assumptions & Dependencies

- Reliable internet connection
- Integration with payment gateways and SMS/email services

3. System Requirements

3.1 Functional Requirements

FR-1: Tenant Management

- FR-1.1: System shall allow admins/managers to add tenants with details: company name, phone, email, address, room number, lease start/end dates etc..
- FR-1.2: System shall allow attachment of lease agreement files (PDF, images)

FR-2: Space Management

- FR-2.1: System shall store room details: number, size (sq. ft.), monthly rent, status (vacant/occupied/under maintenance etc..)
- FR-2.2: System shall display rooms on an interactive floor plan with color-coding for occupancy status

FR-3: Payment Management

- FR-3.1: System shall track rent, utility (electricity/water), and other payments per tenant
- FR-3.2: System shall support manual and automated rent calculation based on room size and discounts
- FR-3.3: System shall accept payments via cash, bank transfer, and mobile banking
- FR-3.4: System shall generate receipts/invoices for payments
- FR-3.5: System shall send automatic reminders (SMS/email) for overdue payments
- FR-3.6: Tenants shall view their payment history and upcoming invoices

FR-4: Maintenance Management

- FR-4.1: Tenants shall submit maintenance requests (e.g., light replacement, water leakage)
- FR-4.2: System shall track request status (received, in progress, resolved)

- FR-4.3: Tenants shall view the status of their requests

FR-5: Communication System

- FR-5.1: System shall provide an internal messaging system between management and tenants

FR-6: Reporting

- FR-6.1: System shall generate reports: monthly income, overdue payments, lease renewals, vacant rooms, tenant list (need to clearly define what are the customer needs)
- FR-6.2: Dashboards shall show: monthly income graph, vacant room count, overdue payment count (need to clearly define what are the customer needs)

FR-7: User Roles & Access Control

- FR-7.1: System shall enforce role-based permissions (Admin, Manager, Accountant, Tenant)
- FR-7.2: Tenants shall only access their own data

3.2 Non-Functional Requirements

- Performance: Support up to 500 concurrent users
- Availability: 99% uptime
- Scalability: Cloud-ready, scalable architecture
- Security: Role-based access control, data encryption
- Usability: Myanmar language, intuitive design
- Compatibility: Web-based (responsive), future mobile support

4. External Interface Requirements

4.1 User Interfaces

- Web portal (React)
- Future mobile apps (iOS/Android)
- Admin, Manager, Accountant, and Tenant dashboards

4.2 Hardware Interfaces

- Cloud servers (AWS)
- Compatible with modern browsers and mobile devices

4.3 Software Interfaces

- Payment gateway APIs
- SMS/Email service APIs
- Document storage services

4.4 Communication Interfaces

- HTTPS/TLS for secure communication
- RESTful APIs for integration

5. Appendices

5.1 Glossary

- KYC: Know Your Customer
- RPO: Recovery Point Objective
- RTO: Recovery Time Objective

5.2 Future Enhancements

- E-Invoicing via email
- Mobile app for management
- Online payment portal for tenants

5.3 References

- Sample SRS for Insurance Management System